THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The speed of light in vacuum is 1

(a) 3 x 10^6 m/s **(b)** 3 x 10^8 m/s **(c)** 3 x 10^10 m/s **(d)** 3 x 10^5 m/s

2. The centre of a lens is called the 1

(a) pole **(b)** optical centre **(c)** aperture **(d)** focus

3. The image in a plane mirror is 1

(a) inverted **(b)** real **(c)** virtual **(d)** diminished

4. The refractive index of vacuum is 1

(a) infinity **(b)** 1 **(c)** 1.5 **(d)** 0

5. The unit of power of a lens is 1

(a) metre **(b)** joule **(c)** pascal **(d)** dioptre

6. A concave mirror is also called 1

(a) flat mirror **(b)** converging mirror

(c) plane mirror **(d)** diverging mirror

7. Refractive index has 1

(a) unit of dioptre **(b)** no unit **(c)** unit of metre **(d)** unit of second

8. The power of a convex lens is 1

(a) zero **(b)** positive **(c)** undefined **(d)** negative

9. A convex mirror is also called 1

(a) dentist's mirror **(b)** plane mirror

(c) converging mirror **(d)** diverging mirror

10. A medium with a larger refractive index is called 1

(a) transparent only **(b)** opaque **(c)** optically denser **(d)** optically rarer

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the relation between focal length and radius of curvature. 2

12. Write the formula for power of a lens. 2

13. What is the pole of a mirror? 2

14. Define refraction of light. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Name the three characteristics of the image formed by a plane mirror. 3

16. A lens has a focal length of 20 cm. Find its power. 3

17. What are the uses of a convex lens? 3

OR

A concave mirror has a focal length of 10 cm. Find its radius of curvature. 3

18. Find the focal length of a lens of power +2 D. 3

OR

Name the type of image formed on a screen. Can a virtual image be formed on a screen? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. State the laws of refraction and write Snell's law. 5

OR

Write the sign conventions used for spherical mirrors. 5

20. Define refractive index. Write its formula. Find the speed of light in water if n = 1.33 and c = 3 x 10^8 m/s (approx.). 5

OR

Explain the power of a lens. Find the power of a lens of focal length -25 cm and +50 cm. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND